Improving Grounded Question Answering with Retrieval-Augmented Generation: A Small-Scale Study

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Abstract

Large language models often hallucinate when asked about content they were not trained on. This paper presents a small-scale study of Retrieval-Augmented Generation (RAG) for document question answering. We combine local sentence-transformer embeddings with a FAISS vector index and a hosted LLM. On a test set of 120 questions across 15 documents, the RAG system achieved 87 percent answer accuracy versus 41 percent for the same model without retrieval, while reducing hallucinated answers from 33 percent to 4 percent.

1. Introduction

Question answering over private documents is a common enterprise need. Fine-tuning is costly and static, while plain prompting suffers from hallucination. RAG addresses both problems by retrieving relevant passages at query time and grounding the model's answer in that evidence. Our study evaluates a minimal RAG pipeline suitable for prototypes and internships.

2. Method

Documents are split into chunks of 1000 characters with 150-character overlap using a recursive splitter. Each chunk is embedded with the all-MiniLM-L6-v2 sentence transformer (384 dimensions) and stored in a FAISS flat index. At query time, the question is embedded and the top-4 chunks by cosine similarity are inserted into a grounded prompt that forbids outside knowledge and requires an explicit refusal when the context is insufficient.

3. Results

The RAG configuration answered 104 of 120 questions correctly (87 percent). The no-retrieval baseline answered 49 of 120 (41 percent). Hallucination, measured as confident answers unsupported by the source documents, dropped from 33 percent to 4 percent. Median end-to-end latency was 2.8 seconds, dominated by LLM generation rather than retrieval, which averaged 21 ms.

4. Limitations

The study covers only text-based documents in English; scanned PDFs would require OCR. The flat FAISS index is appropriate up to roughly one million chunks, beyond which approximate indexes are recommended.

5. Conclusion

Even a minimal RAG pipeline with local embeddings and a small hosted model substantially improves accuracy and nearly eliminates hallucination for document question answering, making it a strong default architecture for assistant products such as document copilots.